# ChatGPT 對話紀錄：Kaggle 多元線性回歸選題

HW2 多元線性回歸 × CRISP-DM | 用途：選擇特徵數介於 14–39 的 Kaggle 資料集
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**使用者**

list kaggle multiple linear regression problems , choose features > 13, < 40

**ChatGPT**

### Kaggle Multiple Linear Regression Problems (14–39 Features)

你希望找適合 Multiple Linear Regression（多元線性回歸）的 Kaggle 題目，條件為：13 < Number of Features < 40，也就是 14–39 個輸入特徵，不包含 Target（預測目標）。

建議優先選擇連續數值預測（Regression）、可使用 sklearn.LinearRegression、適合進行 Feature Selection、MSE、RMSE、$R^2$ 與五大回歸假設檢驗的資料集。

特別注意：Kaggle 上常見的 House Prices — Advanced Regression Techniques 有 79 個原始特徵，不符合你的原始特徵數限制，但可以透過特徵選擇建立 14–39 特徵的練習版本。

### Recommended Kaggle Regression Problems

以下區分「原始特徵符合條件」與「需要 Feature Selection 才符合條件」。

| # | Kaggle Problem | Features | Target | Suitability |
|---|---|---|---|---|
| 1 | Predict the Flood | 20 | FloodProbability | Excellent |
| 2 | Flood Prediction Dataset | 20 | FloodProbability | Excellent |
| 3 | Istanbul Apartment Prices 2026 | 29 or fewer* | Price | Good |
| 4 | Swiss Housing Price Prediction | 30 | House price | Good; restricted access |
| 5 | Human–AI Interaction | 20 | Trust score | Excellent; synthetic |
| 6 | House Prices — Advanced Regression | 79 → select 20–30 | SalePrice | Excellent for feature selection |

*Istanbul 資料集描述為 30 欄，包含價格等欄位；精確輸入特徵數需檢查 CSV。上述資料集的特徵數以來源描述為準，類別編碼後可能增加。

### My Top 3 for Teaching

1. **Predict the Flood**：20 numerical features, continuous target. Best for teaching OLS, MSE, $R^2$, residual analysis and regression assumptions.
2. **Istanbul Apartment Prices**：Suitable for categorical encoding, feature selection, and housing-price regression. Check the dataset columns before modeling.

3. **Human–AI Interaction**：20 original features, 10,100 samples. Useful for studying coefficients, categorical features and multicollinearity.

**Suggested Assignment: CRISP-DM + Feature Selection**
For any selected dataset, students can complete:

- Business Understanding: Define the continuous prediction target.
- Data Understanding: Verify sample size, feature count, missing values and correlations.
- Data Preparation: Train/test split, missing-value handling and categorical encoding.
- Modeling: Train LinearRegression, Ridge and Lasso.
- Evaluation: Compare MAE, MSE, RMSE, $R^2$ and Adjusted $R^2$; check the five regression assumptions.
- Deployment: Predict new samples and explain the model.

For Feature Selection, compare SelectKBest, RFE, Sequential Feature Selection, Lasso and permutation importance, selecting 14, 20, 25 or 30 features using training data only.

My first choice is Predict the Flood, because its 20 numerical features meet your constraint directly and avoid the extra complexity of categorical encoding.